Piter Z. Garcia Bautista

NSF Noyce Scholar — Data Science & Bioinformatics Graduate Student

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Research Summary

NSF Robert Noyce Scholar and PhD-track graduate student specializing in **equitable bioinformatics** and **fairness-aware machine learning** for healthcare and educational applications. Developer of an interdisciplinary "EQUITAS" framework that integrates computational rigor, educational equity, and lived experience to address diagnostic disparities. Seeking Graduate Assistantship opportunities to advance research in ethical AI, algorithmic fairness, and translational bioinformatics while supporting faculty research and teaching initiatives.

Education

- **Master of Science in Data Science** — Expected Aug 2026
  Rochester Institute of Technology, Golisano College of Computing — *Rochester, NY*
- Concentration: Bioinformatics & Artificial Intelligence
- Relevant Coursework: DSCI-633 (Foundations of Data Science), DSCI-644 (Software Engineering for Data Science), DSCI-601 (Applied Data Science I), BIOL-630 (Bioinformatics Algorithms), ISTE-780 (Data-Driven Knowledge Discovery)
- Current GPA: 3.9+ (Excellent Academic Standing)
- **Master of Science in Teaching Computer Science K-12** — Expected Aug 2026
  University of Rochester, Warner School of Education — *Rochester, NY*
- NSF Robert Noyce Teacher Scholarship Program (Full funding: $30,000)
- Advanced Certificate in Leadership in Disability and Inclusive Practices
- Relevant Coursework: ED-447 (Disability and Schools), ED-405 (Assessment in Instructional Contexts), ED-452 (Instructional Strategies for Inclusive Classrooms), EDE-498 (Integrating CS Standards)
- Service Commitment: 2 years teaching in high-need school districts
- **Master of Science in Computer Science** — 2015
  Rochester Institute of Technology, Golisano College of Computing — *Rochester, NY*
- Concentrations: Computer Graphics, Artificial Intelligence, Big Data Analytics
- Project: Technology for Automatic Sign Language Interpreter
- Research: Big Data Medical Diagnosis: A Multi-Method Approach
- **Bachelor of Science in Computer Engineering** — 2012
  Farmingdale State College — *Farmingdale, NY*
- Dual Degree: Electrical Engineering & Computer Engineering Technology
- Project: Smart Meter Home Application
- Honors: CSTEP Award & Merit, Dual BS Scholarship

Current Research Experience & Projects

- **Lead Researcher - EQUITAS Framework for Diagnostic Justice** — 2024 - Present
  Independent Research Initiative, RIT & UOR — *Rochester, NY*
- Developed interdisciplinary research framework integrating bioinformatics, educational equity, and trauma-informed care
- Published "Trauma-Informed Bioinformatics for Diagnostic Justice" establishing a systematic approach to bias detection
- Demonstrated 42% reduction in misdiagnosis rates for marginalized populations through equity-aware algorithms
- Collaborated with cross-institutional teams spanning computer science, education, and health sciences
- **Principal Investigator - Equitable Bioinformatics Project (ISTE-780)** — 2024 - Present
  Data Science Program, RIT — *Rochester, NY*
- Pioneered first systematic application of fairness assessment to RNA secondary structure prediction

- Implemented comprehensive bias detection framework revealing critical equity violations (11.1% equity score)
- Developed modular codebase (900+ lines) enabling reproducible equity-aware algorithm evaluation
- Achieved statistical significance ($p < 0.01$) in algorithmic bias detection across demographic groups

- **Research Developer - RNA Structure Prediction & Visualization (BIO-630)** 2024
  Bioinformatics Program, RIT *Rochester, NY*
- Advanced single-sequence RNA structure prediction integrating Nussinov algorithm with MFE enhancements
- Integrated dynamic programming with thermodynamic modeling using Turner parameters and ViennaRNA
- Achieved >90% accuracy on synthetic motifs with comprehensive benchmarking against biological RNAs
- Created validated, open-source codebase for high-throughput feature extraction and analysis

Teaching & Mentoring Experience

- **STEM Tutor & Academic Mentor** 2021 - Present
  Varsity Tutors & Independent Practice *Remote & In-Person*
- Deliver comprehensive tutoring in Computer Science, Data Science, Mathematics, and Statistics
- Develop culturally responsive lesson plans supporting diverse learning needs and disabilities
- Mentor underrepresented students in STEM fields, with a focus on neurodivergent learners
- Maintain 95% student satisfaction rate with demonstrated improvement in academic performance

- **Faculty Assistant - Graduate Office** 2013 - 2015
  Rochester Institute of Technology, Golisano College of Computing *Rochester, NY*
- Assisted with all administrative tasks related to graduate student services and program operations
- Supported faculty and staff with graduate program coordination and student support initiatives
- Managed documentation, communications, and administrative processes for graduate programs
- Provided direct assistance to graduate students with program requirements and procedures

- **Private Tutor - Computer Science** 2013 - 2015
  Independent Practice *Rochester, NY*
- Delivered comprehensive tutoring in Java programming and Computer Theory and Foundations
- Developed personalized lesson plans for graduate and undergraduate computer science students
- Assisted students with complex programming concepts, algorithms, and theoretical foundations
- Maintained high success rate in helping students improve academic performance and understanding

Latest Publications & Academic Contributions

- **Garcia, P.** (2025). Equitable Bioinformatics: Enhancing Diagnostic Decision-Making through RNA and Biomarker Data. *ISTE-780 Project Report*, Rochester Institute of Technology.
- **Garcia, P.** (2025). Trauma-Informed Bioinformatics for Diagnostic Justice: An EQUITAS Framework for Integrative Healthcare and Educational Systems. *ED-447 Capstone Project*, University of Rochester.
- **Garcia, P.** (2025). RNA Secondary Structure Prediction and Visualization: Implementing the Nussinov Algorithm with MFE Enhancements. *BIO-630 Final Project*, Rochester Institute of Technology.
- **Garcia, P.** (2025). Navigating Behavior and Special Education: A First-Person Reflection on Diagnostic Equity. *Policy Analysis Paper*, University of Rochester.
- **Garcia, P.** (2015). ASL Recognition Application: Image Processing for Sign Language Translation. *MS Thesis*, Rochester Institute of Technology.
- **Garcia, P.** (2014). "Big Data Medical Diagnosis: A Three-Method Approach to Improve Diagnostic Accuracy." *Conference Presentation*, RIT Graduate Research Symposium.

Latest Technical Skills

Programming Languages: Python, R, Java, C++, C#, MATLAB, SQL, JavaScript, HTML/CSS, Bash
Bioinformatics & Computational Biology: ViennaRNA, Nussinov Algorithm, Turner Parameters, BLAST, SHAP Analysis
Machine Learning & AI: TensorFlow, Keras, scikit-learn, XGBoost, PyTorch, LSTM, Fairness-aware ML
Data Science & Analytics: Pandas, NumPy, SciPy, Matplotlib, Plotly, Seaborn, Statistical Analysis, SPSS
Cloud & Infrastructure: AWS, GCP, Azure, Docker, Git/GitHub, Linux/Unix Systems
Databases: PostgreSQL, MySQL, MongoDB, NoSQL
Specialized Frameworks: EQUITAS Fairness Assessment, Bias Detection, Disparate Impact Analysis

Latest Professional Experience

- **Data Solutions Engineer** — Sep 2022 - Aug 2024
  VEDADATA Inc. — *New York, NY*
- Designed and implemented data-driven solutions using Python, Pandas, and cloud technologies
- Architected scalable solutions addressing complex business problems with an emphasis on accessibility
- Collaborated with cross-functional teams to deliver robust, maintainable software systems
- **AI Data Solutions Engineer** — Jan 2021 - Sep 2022
  VIOME Inc. — *New York, NY*
- Developed AI-driven solutions for healthcare applications with a focus on equitable outcomes
- Implemented best practices for AWS deployment and data management with SDLC compliance
- Led development of predictive models for personalized health recommendations
- **Computer & Data Science Consultant** — Mar 2017 - Jan 2021
  Samasta Health Foundation — *Remote & Optional*
- Provide pro-bono consulting to architect smart systems supporting children in need
- Develop AI-driven healthcare solutions with an emphasis on equity and accessibility
- Collaborate with healthcare professionals to implement trauma-informed technology solutions

Relevant Awards & Recognition

- **NSF Robert Noyce Teacher Scholarship** (2024-2026) - Full funding for MS in Teaching CS K-12
- **MS International Scholarship** (2012-2015) - MESCYT & RIT - Academic excellence recognition
- **Honorable Mention Technology Award** (2012) - NYS STEP/CSTEP - 2nd place capstone project
- **IEEE Alumni Member** (2009-Present) - Member No. 90613000
- **Academic Excellence Merit** (2002-2006) - Mateo Lara High School

Faculty Collaboration Interests

I am particularly interested in collaborating with faculty whose research aligns with my interdisciplinary approach:

- **Dr. Qi Yu** - ISTE-780 Professor, whose human-centered AI and machine learning approaches directly inspire my equitable bioinformatics project development
- **Dr. Fernando Rodriguez** - BIO-630 professor, whose expertise in comparative genomics and molecular evolution provided foundational guidance for my RNA secondary structure enhancement work
- **Dr. Travis Desell** - DSCI-640 Professor, Faculty advisor, and Data Science MS Program Coordinator, whose neuroevolution and distributed computing research has inspired my algorithmic approach
- **Dr. Feng Cui** - Bioinformatics Graduate Director, whose pioneering work in computational biology and AI applications in medicine aligns perfectly with my health equity focus
- **Dr. Evan Selinger** - AI ethics authority whose research on algorithmic accountability and responsible AI development provides the ethical foundation essential to my EQUITAS framework
- **Dr. Christian Newman** - Graduate Program Director whose empirical research in software engineering and educational technology supports both my technical development and teaching aspirations

References, detailed CV or portfolio, code or paper samples, are available upon request